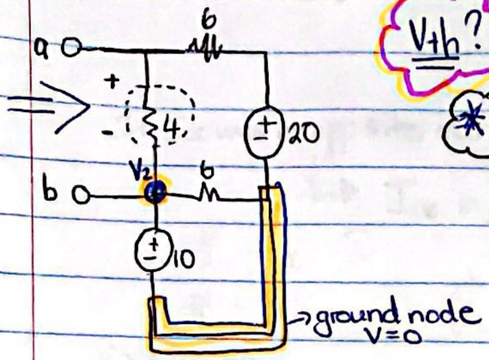
16 April
EBN :



Vth?
- need voltage of 4 Ω

* method : Nodal
* But you know that $V_2 = 10V$

$$\frac{V_1 - 20}{6} + \frac{V_1 - 10}{4} = 0$$

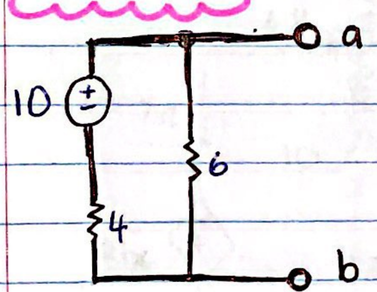
$$V_1 = 14V$$

But it is a imaginary supernode.

$$V_1 - V_2 = V_{ab} = V_{th}$$

$$14 - 10 = 4V = V_{th}.$$

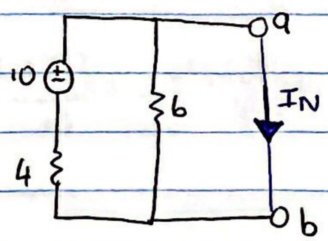
Norton :



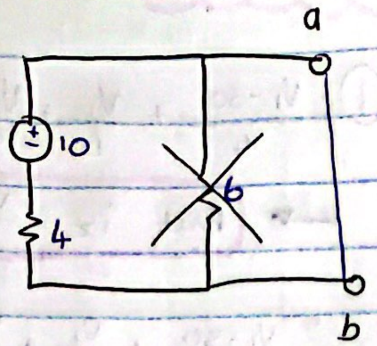
① 1 step: close a gap with a line
② Assign current moving from a to b.

Norton : I_N

→ we will use nodal or KVL (only when you have one box.)



- Doing with respect to a-b. So current is different than from Rth



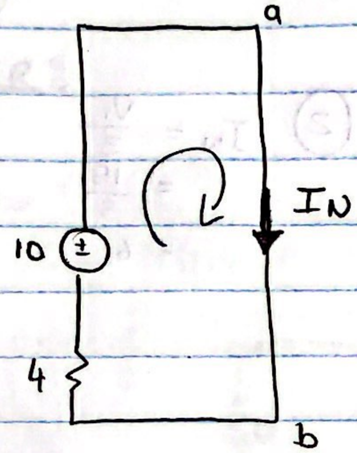
→ Remove resistor as $V=0$

as

$$I = \frac{V}{R}$$

$$0 = \frac{V}{R}$$

$$0 = V$$

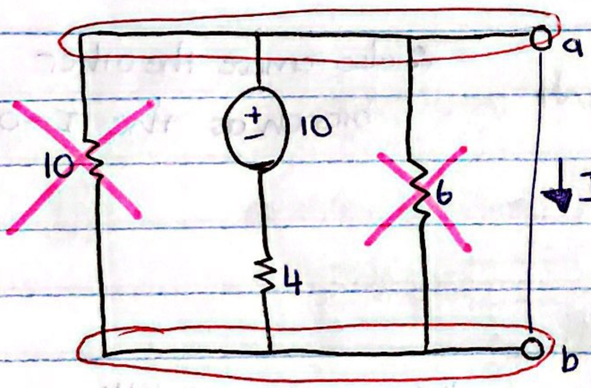


Do KVL:

$$4 I_N = 10$$

$$I_N = 2.5$$

Exz:

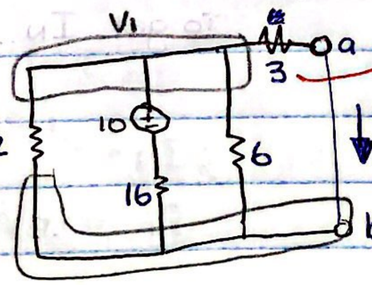


→ when you close a line: check if you have parallel resistors

→ check nodes then you will see the parallel resistors.

Then do KVL

Ex3:



→ you can't just cancel.

∴

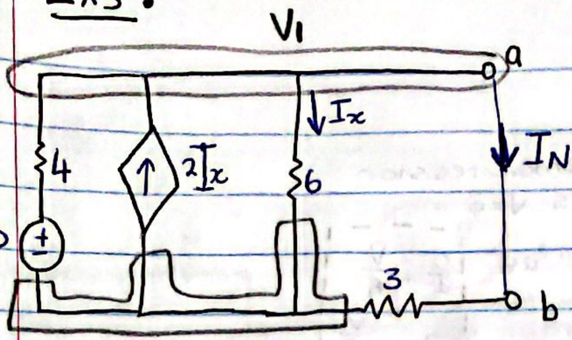
$$\frac{V_1}{12} + \frac{V_1 - 10}{16} + \frac{V_1}{6} + \frac{V_1}{3} = 0$$

$V_1 = \dots$

So then $I_N = \frac{V_1}{3}$

Ex3 :

Calculate I_N.



① $\frac{V_1 - 30}{4} + \frac{V_1}{6} + \frac{V_1}{3} - 2i_x = 0$

→ $\boxed{\text{But } i_x = \frac{V_1}{6}}$

$$\frac{V_1 - 30}{4} + \frac{V_1}{6} + \frac{V_1}{3} - 2\left(\frac{V_1}{6}\right) = 0$$

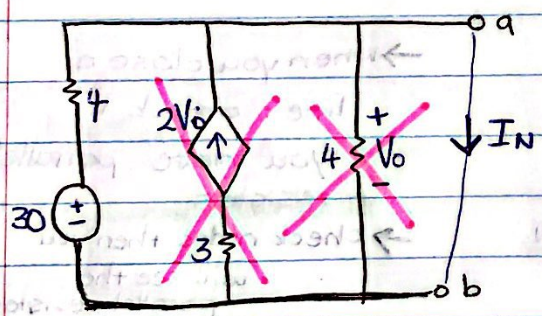
$$V_1 = 18\,V$$

② $I_N = \frac{V_1}{3}$

$= \frac{18}{3}$

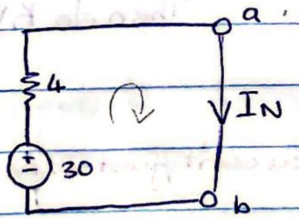
$= 6$

Ex 4 :



→ $V_o = 0$ because of the parallel.

∴ also remove the other branch as the $I = 0$



Then you can use KVL
To get I_N...

$4 I_N = 30$

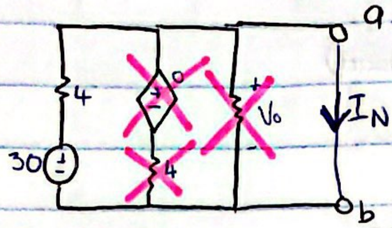
$I_N = 7.5\,A$

Scanned with CamScanner

Ex 5 :

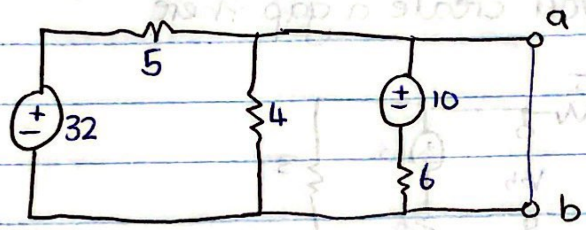
→ $V_o = 0$

→ then the 4Ω also cancels out as the 4Ω is in parallel again...



MAX POWER:

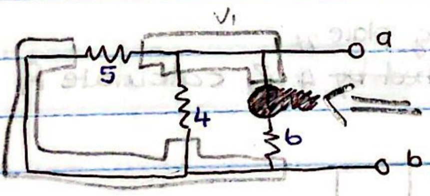
$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$



Make sure that you have a gap → Vth

Rth :

① close with line



② use $\frac{1}{r_p} = \frac{1}{5} + \frac{1}{4} + \frac{1}{6}$

$r_p = 1.62\ \Omega$

Vth

$$\frac{V_1 - 32}{5} + \frac{V_1}{4} + \frac{V_1 - 10}{6} = 0$$

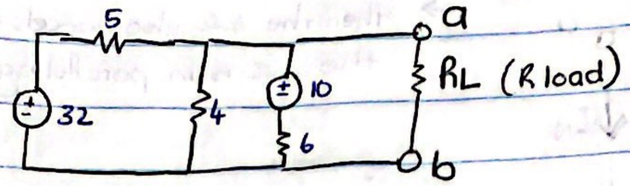
$V_1 = 13.08$

[redacted]

→ $P_{max} = \frac{(13.08)^2}{4(1.62)}$

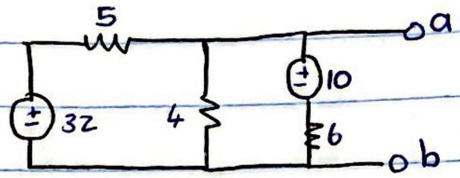
$= \ldots$

when they ask for R load:



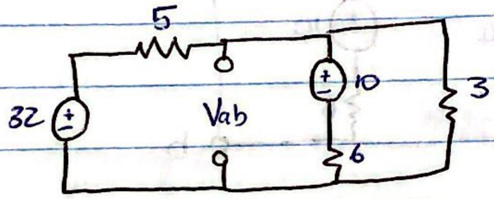
Calculate Pmax absorbed by Rload

You should make a gap:



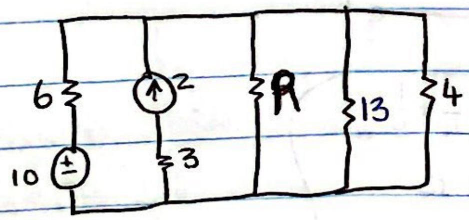
when you calculate Vth they will always give you a gap.

OR if they say at 4 Ω for example then you create a gap there



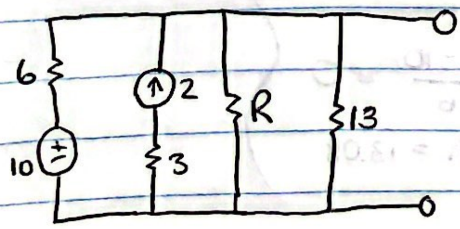
$$\frac{V_{th} - 32}{5} + \frac{V_{th} - 10}{6} + \frac{V_{th}}{3} = 0$$

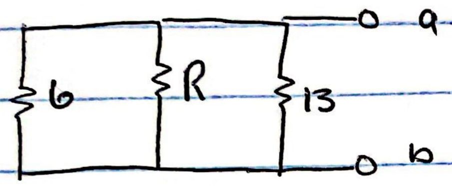
Ex 6 Pmax taking place at ~~absorbed by~~ 4 Ω calculate R



∴ Rth is 4 Ω because they say Pmax is taking place there.

→ go on as if calculating Rth.





$$\left(\frac{1}{6}+\frac{1}{R}+\frac{1}{13}\right)^{-1} = 4$$

$$R = 156\ \Omega$$